Aditya Saini

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Education

Purdue University West Lafayette, IN
B.S. Computer Science (Systems Programming) *Expected Graduation: May 2028*
Relevant Coursework: Linear Algebra, Multivariate Calculus, Data Structures & Algorithms, Discrete Mathematics, Computer Architecture, Programming in C, Object-Oriented Programming

Experience

Robotic Safety Research Assistant January 2026 – Present
PurSec, Purdue Computer Science *West Lafayette, IN*

- Researched safety of multimodal LLMs controlling embodied agents, formulating attack methods to create physical harm from benign individual steps for over 20 individual attacks spanning 8 different physical attack categories
- Created a dual-brain verification method to measure defensive success, consisting of RoboGuard LTL verification and the ASIMOV constitution checks running on GPT-4o to allow for quicker verifications for each attack
- Built a closed-loop harness through a state machine in Python that returns updated scene state and full action history to the verifier after every robot step, gating each proposed action on both safety brains before execution
- Built the Python simulation/eval pipeline using AI2-THOR rendering, Pillow + FFmpeg video output, and a Dockerized Ubuntu 22.04 + Xvfb headless harness for reproducible runs on machines with or without displays

Embedded Engineer Sep. 2025 – Present
Purdue Electric Racing *West Lafayette, IN*

- Designed and implemented the driver-facing dashboard UI for the Electric Racing competition vehicle on a Nextion HMI display using Nextion Editor, authoring all screen graphics for live telemetry, fault codes, and sensor data
- Wrote embedded C on an STM32F4 (ARM Cortex-M4) driving 6+ dashboard screens and button-mapped page transitions over UART, mapping CAN fault codes to prioritized driver alerts
- Rewrote the display command queue from fixed-width slots to a packed byte-level ring buffer, eliminating padding on the UART stream and fixing intermittent dropped commands on the Nextion dashboard display

Build President & Programming Captain September 2021 – August 2025
FRC Team 2554 - The Warhawks *Edison, NJ*

- Led 60+ students as Build President to District Championships after 18 years, earning 3 regional awards.
- Built a CUDA-accelerated OpenCV vision pipeline in Python for AprilTag detection, improving autonomous positioning accuracy by 66%, leading to a larger scoring volume with greater scoring efficiency.
- Fabricated a custom 30-button robot controller using a PSoC microcontroller, writing custom embedded C firmware for continuous input polling to ensure reliable driver input under competition conditions
- Diagnosed and resolved cross-system hardware failures across CAN bus-chained motor controllers and sensors
- Designed robot subsystems in Onshape through successive prototyping cycles, including the intake and drivetrain

Projects

Memory Copier | *SystemVerilog, RTL Design, Verification* July 2026

- Designed a byte-addressed memory copy controller (simple DMA) as a five-state Moore FSM that reads from a source address and writes to a destination address when a start signal goes high
- Split combinational next-state and output logic from a clocked state register with asynchronous reset, and buffered read data in a register so it held stable across the read-to-write cycle
- Verified copy correctness, memory handshake timing, and reset behavior with a SystemVerilog testbench

Aptum | *TypeScript, Next.js, React, Supabase, OpenAI, KaTeX* July 2026 – Present

- Built an on-demand learning platform that generates a full course on any topic from a learner profile, with structured modules, long-form LaTeX notes, and MCQ practice flashcards per chapter
- Generated syllabi with OpenAI structured JSON schemas scaled to depth and duration, then bulk-wrote LaTeX notes and quizzes for every chapter so the entire course is ready up front
- Cached identical syllabi by SHA-256 profile hash, stored per-user courses in Supabase with RLS, and repaired JSON-corrupted LaTeX so chapter notes render math cleanly with KaTeX

gcstats | *Python, SQLite, HTML* August 2026 – Present

- Built a local CLI that snapshots chat.db, decodes iMessage bodies, and nets out removed tapbacks using stdlib
- Shipped an HTML dashboard of per-capita reaction rates, leaderboards, and who-reacts-to-who tapback matrices

Tava | *Python, TypeScript, OpenAI, Firebase, React Native* September 2025

- Built AI-powered search and swipe-based matching to recommend collaborators for student project ideas, ranking candidates by shared interests via LLM-assisted retrieval. Includes an interactive, visual node-edge graph for direct and secondary connections.

Natural Language Shell | *C, POSIX, Python, Bash, OpenRouter, OpenAI API* June 2025 – January 2026

- Built NLSH, a custom Unix shell in C with process management, tokenization, and built-ins (cd, help, exit)
- Added natural-language command translation via Python: phrase mappings for common queries, LLM fallback through OpenRouter/DeepSeek when no match exists
- Bridged C and Python with popen, OS-aware prompts, and a Bash installer that puts the shell on PATH

Technical Skills

Languages: Python, C, C++, Java, SQL, TypeScript, JavaScript, SystemVerilog, Verilog, RISC-V assembly
Libraries/APIs: OpenAI API, OpenCV, Firebase, Expo, React Native, Mapbox
Tools: Git/GitHub, Linux, CUDA, Docker, AWS EC2, Makefiles, GCC, Cursor, Claude Code
Hardware/EDA: Jetson Orin Nano, STM32, STM32CubeIDE, Cadence Genus, Cadence Innovus, OpenLane, LTspice, FPGA synthesis & timing, RTL verification